Urban GIS Portfolio Project 2

GIS-Based Spatial Analysis for Urban Regeneration and Livability Assessment
Case Study: District 14, Tehran, Iran

Samira Heydari | M.Sc. in Geography and Urban Planning | Tarbiat Modares University, Iran

Software & Methods
ArcGIS | AHP | COPRAS | GIS Spatial Analysis | Microsoft Excel

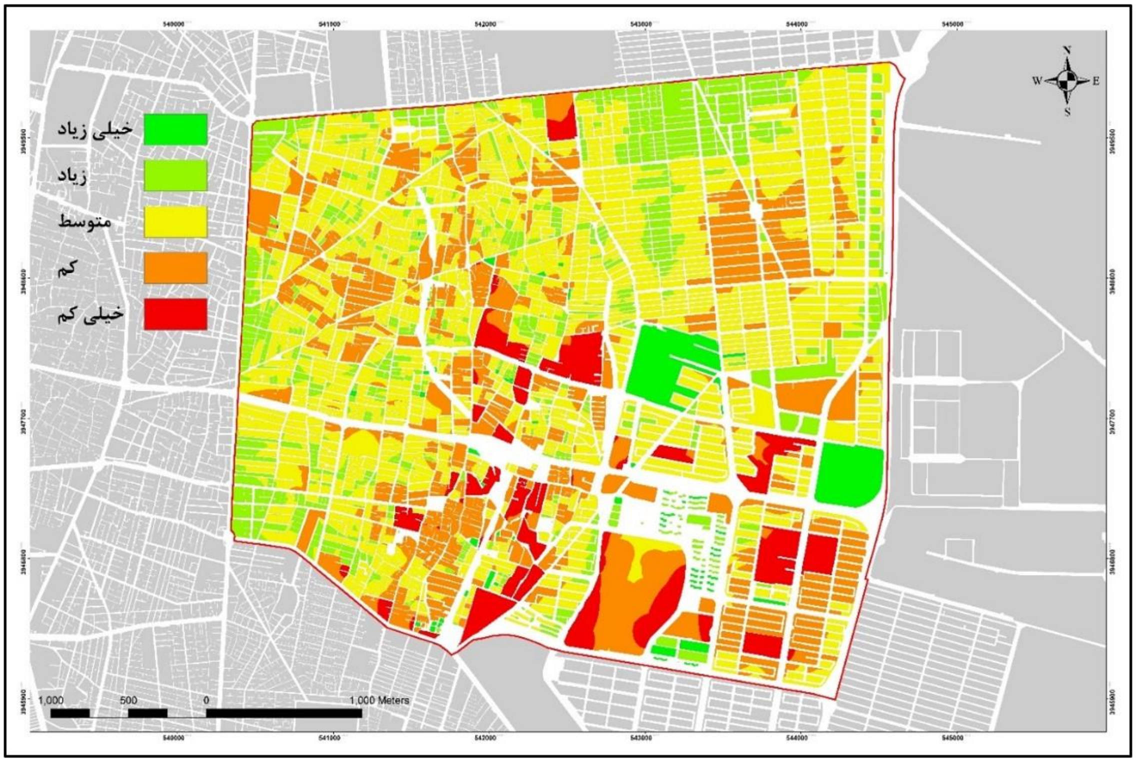

Figure 1. Spatial Zoning of Physical Livability in District 14, Tehran

PROJECT OVERVIEW

This project evaluates the spatial distribution of urban livability within deteriorated neighborhoods of District 14, Tehran. The study integrates Multi-Criteria Decision Making (MCDM) methods with GIS-based spatial analysis to identify priority areas for urban regeneration.

The assessment combines physical, environmental, accessibility, and socio-economic indicators to support evidence-based urban planning and sustainable development.

PROJECT OBJECTIVES

• Evaluate urban livability conditions.
• Identify priority neighborhoods for regeneration.
• Integrate AHP and COPRAS.

- Produce GIS-based livability zoning maps.
- Support sustainable urban planning.

STUDY AREA

District 14, Tehran, Iran

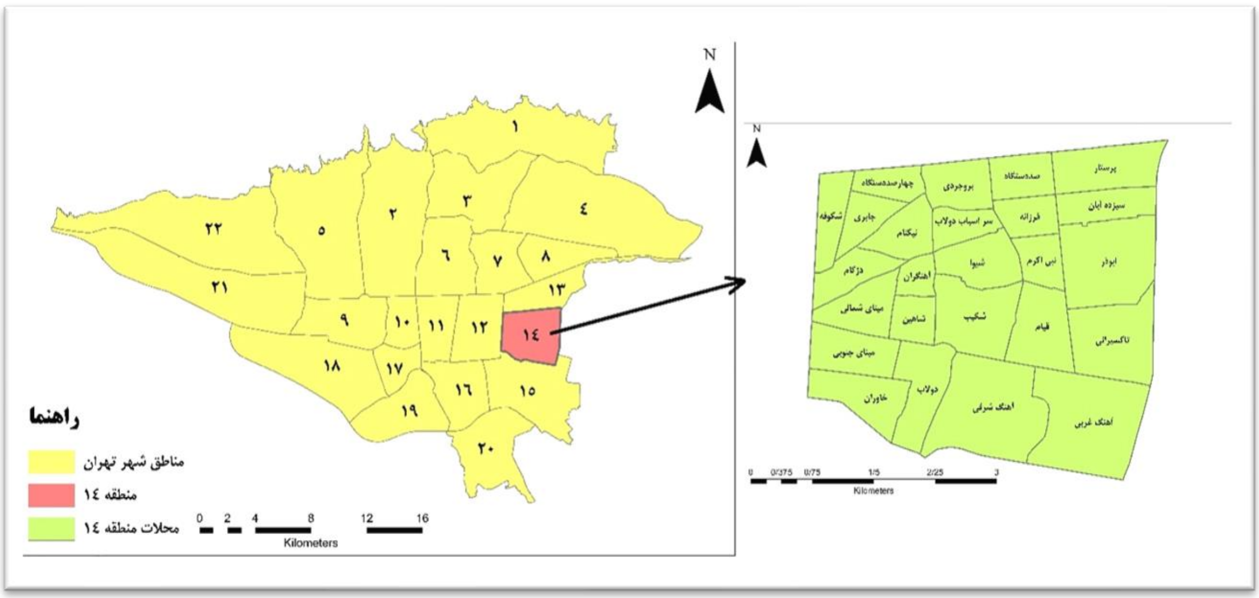

Figure 2. Location of the Study Area: District 14, Tehran

RESEARCH METHODOLOGY

Data Collection → Selection of 13 Indicators → AHP Weighting → COPRAS Ranking → GIS Spatial Analysis → Livability Zoning

Key Skills Demonstrated

ArcGIS
Microsoft Excel
AHP
COPRAS
GIS Spatial Analysis

SPATIAL ANALYSIS

Urban GIS Portfolio Project 2

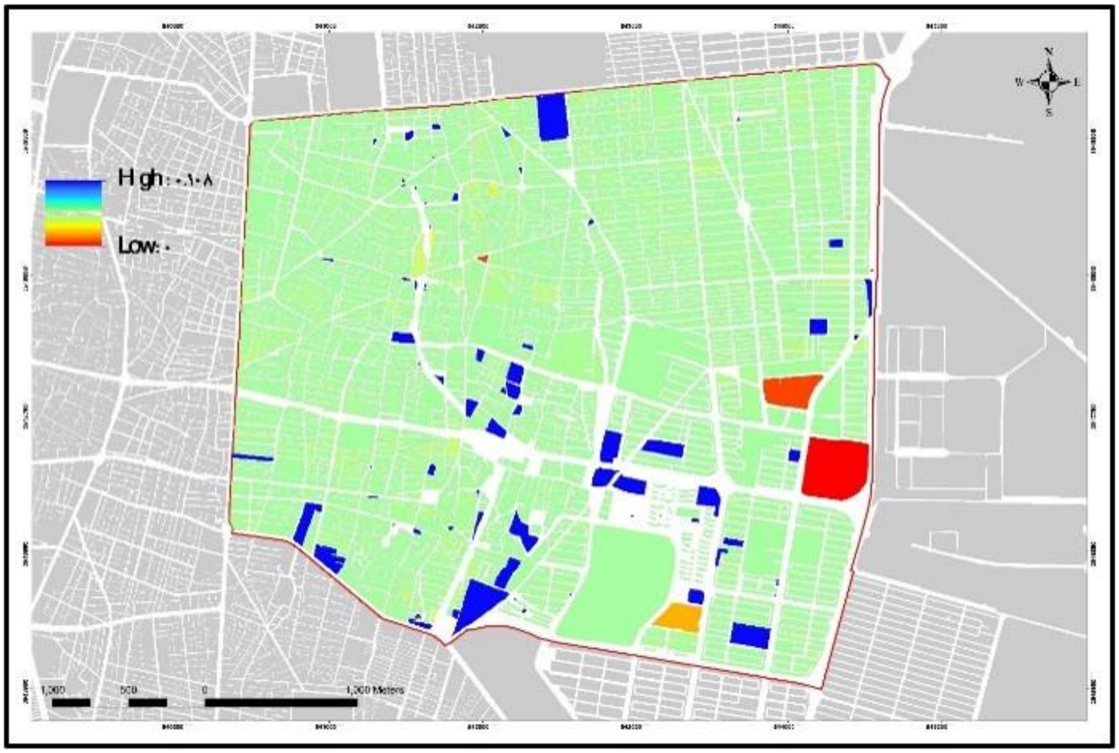

Figure 2. Weighted Standardized Household Density Map

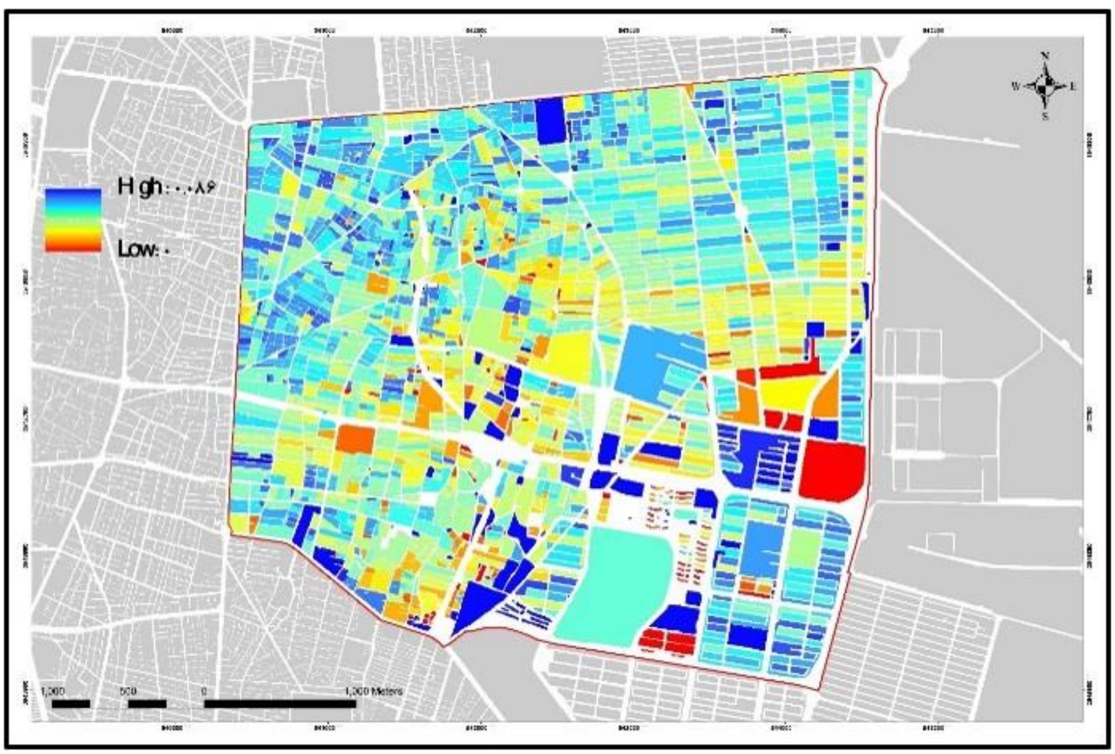

Figure 3. Weighted Standardized Building Age (>30 Years) Map

Urban GIS Portfolio Project 2

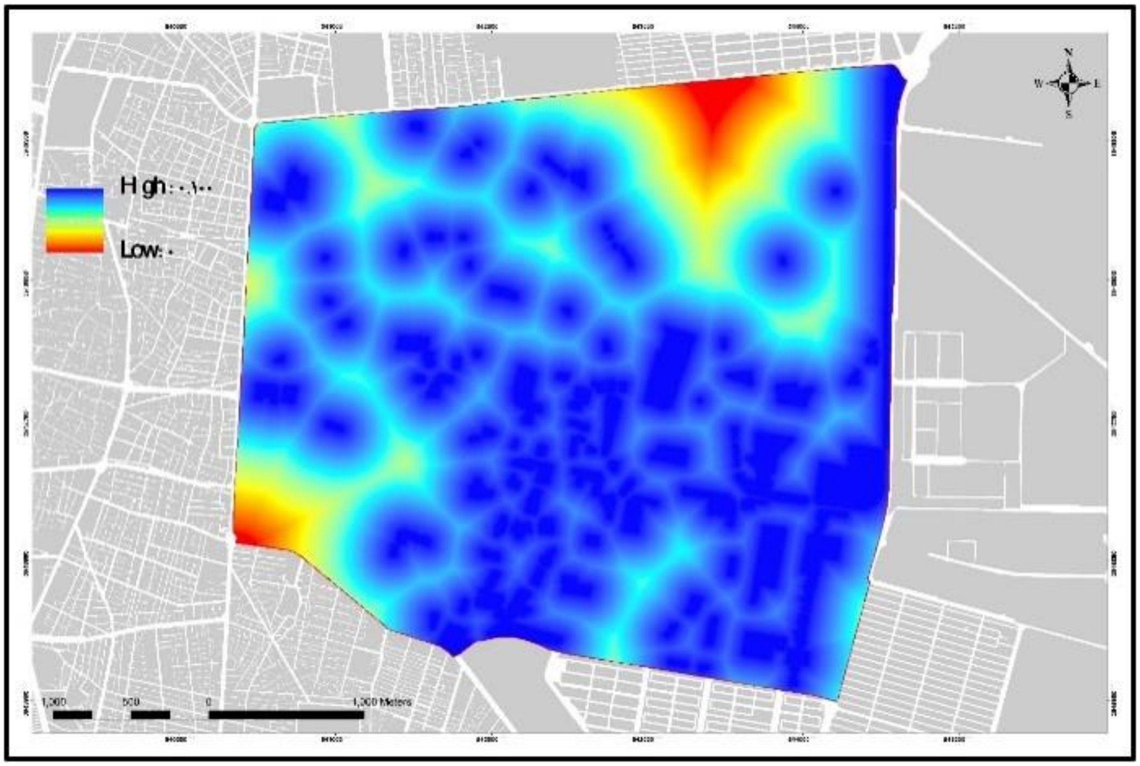

Figure 4. Weighted Standardized Park and Green Space Accessibility Map

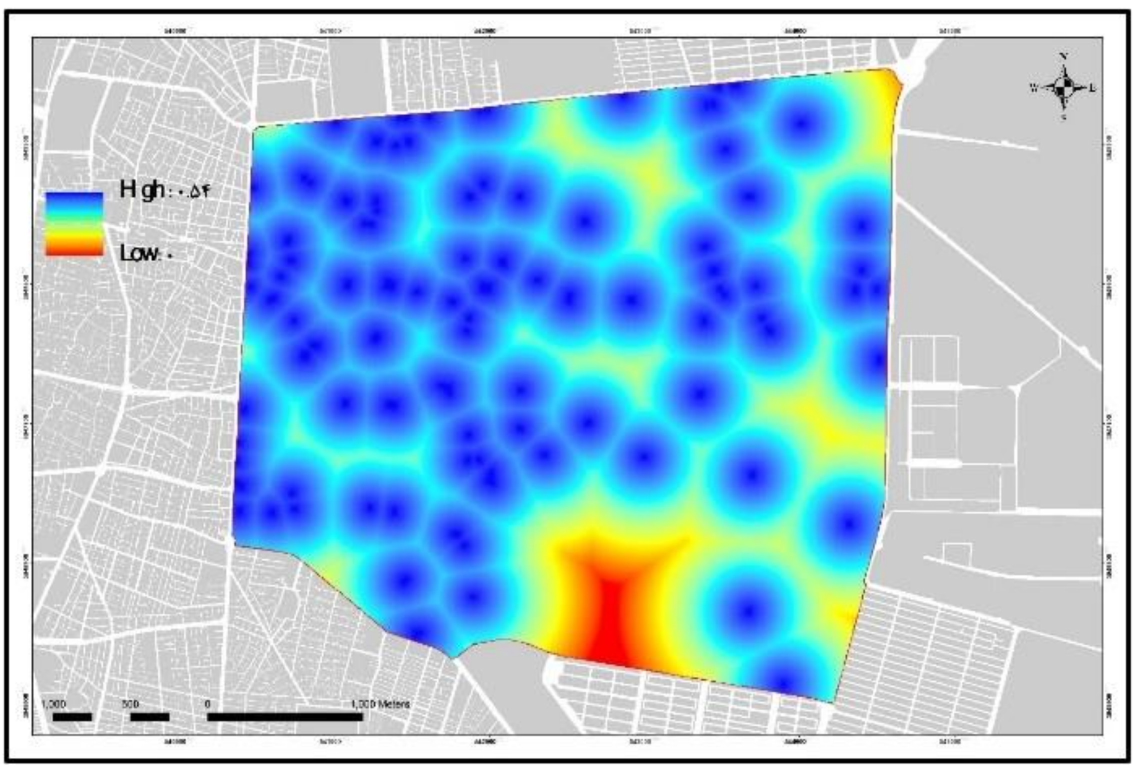

Figure 5. Weighted Standardized Healthcare Accessibility Map

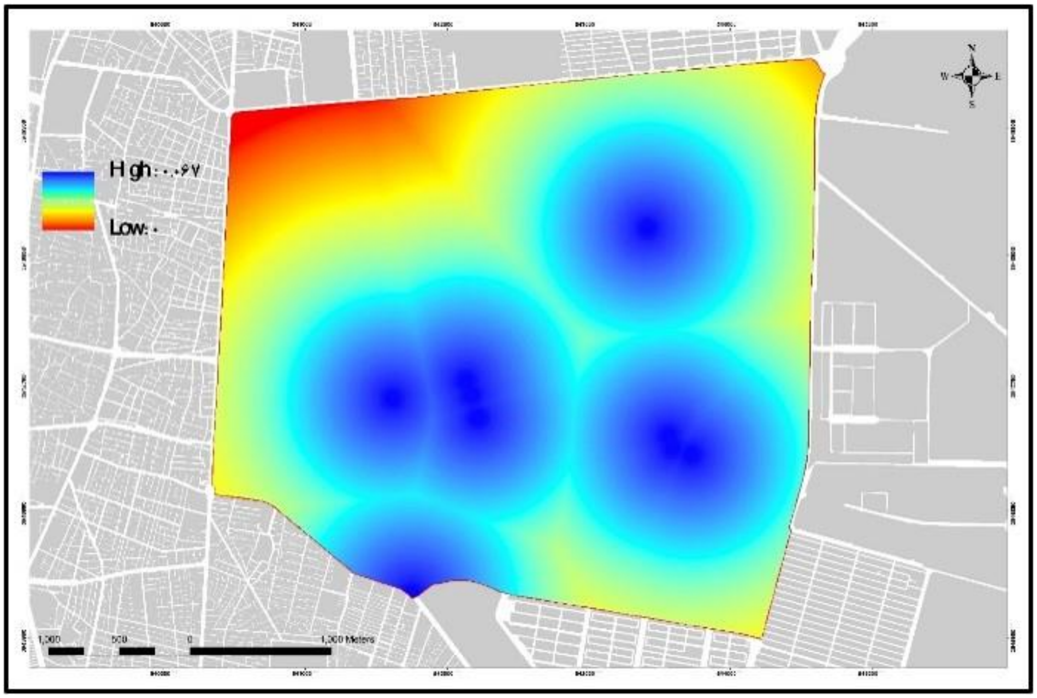

Figure 6. Weighted Standardized Metro Accessibility Map

FINAL LIVABILITY MAP

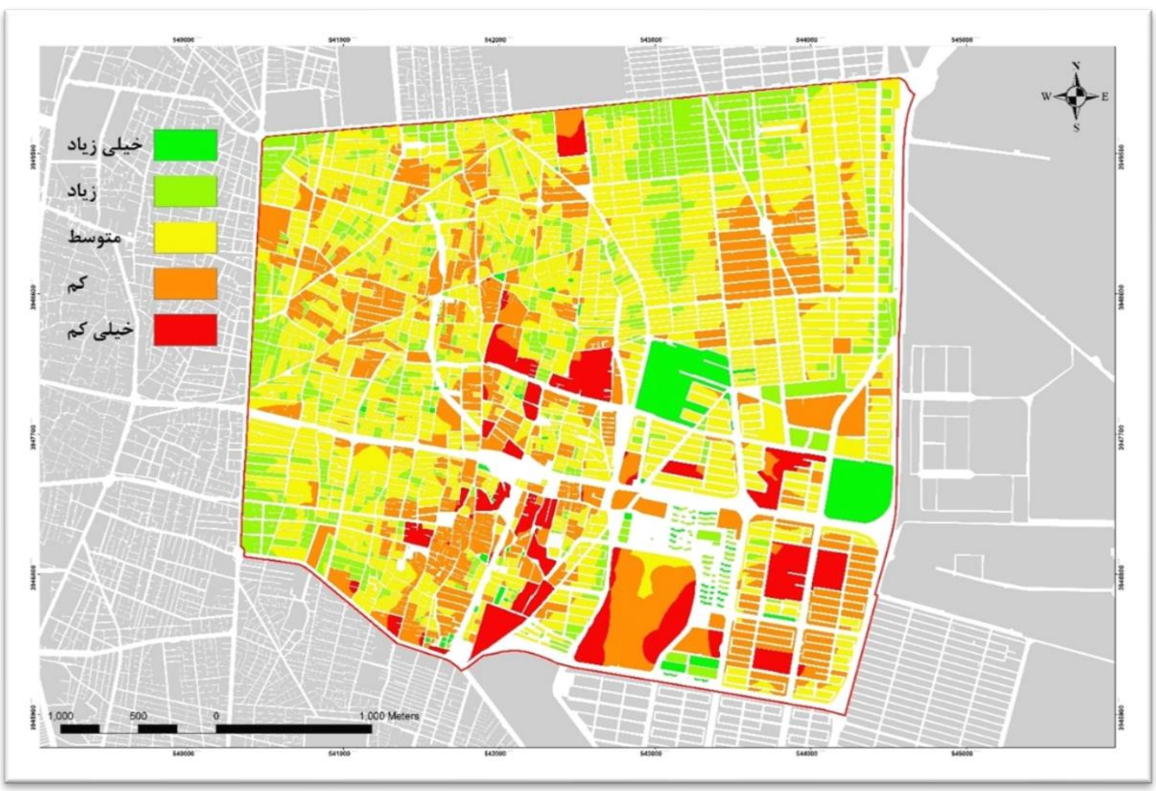

Figure 1. Spatial Zoning of Physical Livability in District 14, Tehran

Urban GIS Portfolio Project 2

KEY FINDINGS

Neighborhoods with better accessibility to green spaces, public transportation, and lower household density achieved higher livability scores, while several deteriorated neighborhoods require priority urban regeneration interventions.

PROJECT OUTCOMES

- Urban Livability Assessment
- GIS Spatial Analysis
- AHP Weighting
- COPRAS Ranking
- Urban Regeneration Planning

PROFESSIONAL SKILLS

- GIS
- Spatial Analysis
- Cartography
- Urban Planning
- Data Visualization